CSTBS v5.1

University Accounts & Free Reservations

VERSION 5.1 / CHANGE REPORT / 01 NOVEMBER 2026

Release summary

Version 5.1 builds on the full Campus Stadium Ticket Booking System v5.0 and makes student and administrator onboarding manageable through the website. It introduces required Student IDs, one-time administrator registration invitations, and fully free university seat reservations. The prior event, seat, booking, cancellation, notification and reporting modules remain in place.

What was added

AREA	IMPLEMENTATION
Student sign-up	Unique 6-20 character Student ID on student registration; stored uppercase, with server-side and database checks.
Admin sign-up	Dedicated /register/admin page accepting a one-use secret key, name, email and password.
First admin key	Secure random key printed once at local application startup when no admin exists; expires after 60 minutes.
Admin invitations	Admin dashboard control to generate another one-time key; private one-time display, one-hour validity.
Student ID updates	Students can complete legacy profiles; admins can correct Student IDs from the website.
Free university seats	No payment collection. Event CRUD offers no price input; all new booking snapshots cost CNY 0.00.
Reports and notifications	Booking and event exports and booking emails describe free CNY admission, not revenue.

Security, migration and behavior

Administrator invitation security

Keys use cryptographic randomness; only keyed HMAC-SHA256 hashes are stored. The registration write and guarded key consumption share one transaction. Expired or replayed keys cannot issue another administrator account.

Who can issue administrator keys

The first local key appears in the private development terminal. After an administrator exists, logged-in administrators issue keys at Admin dashboard > Users. The generated key is shown once in a no-cache page; role and CSRF authorization are required.

Student identity

New student registrations require a unique Student ID. Existing student accounts from V5 remain accessible and can complete their profiles. Administrators can edit IDs without deleting account or booking records.

Free admission and currency

RMB (CNY) is the display currency, but admission is always free. Event price fields are set to zero during schema upgrade; all new booking rows save zero-cost snapshots. Historic booking price snapshots remain for compatibility, are not charged, and are not presented as revenue.

Database migration

V5.1 adds users.student_id, a unique index on non-null IDs, and the admin_invitations table. The upgrade preserves Users, Seats, Events and Booking history. Back up the SQLite database before upgrading, and use the explicit init-db command in production.

Retained modules

V5 authentication, roles, seat blocking, section management, polling availability, booking confirmation and cancellation, Flask-Mail notifications, PDF/CSV exports and administrator dashboard are retained.

Known operational limits

Development mode prints the initial key into the local terminal only. Production provisioning requires a private operator command (reset-bootstrap-key), not logging secrets during worker startup. SQLite, 5-second seat polling, synchronous email and compatibility migrations are appropriate for a small campus pilot and require additional hardening for high-scale production.

Windows setup and GitHub handoff

01 Extract and install

Unzip CSTBS-v5.1.zip; open PowerShell in the project directory. Create .venv, activate it, and install requirements.txt.

02 Configure secrets

Copy .env.example to .env. Generate a random SECRET_KEY locally, paste it into .env and never commit the file.

03 Initialize demo data

Run: python -m flask --app run.py init-db; python -m flask --app run.py seed-events.

04 Run application

Run python run.py and open http://127.0.0.1:5000. Copy the one-time admin key displayed in PowerShell.

05 Create admin and students

Admin uses /register/admin with the key; students use /register with mandatory Student ID.

06 Test before publishing

Run python -m unittest discover -s tests -v, then verify the site, bookings, admin invitations, reports and email behavior locally.

GitHub release information

Tag: v5.1
Title: CSTBS v5.1 - University Accounts & Free Reservations
Notes: docs/github-release-v5.1.md

Verification and limitations

Offline checks performed: Python syntax compilation, Jinja2 template compilation, JavaScript syntax checks and SQLite constraint/migration verification. Live Flask integration tests could not be executed here because the Flask dependencies were not available from the package source; execute them on your local computer before deployment.